numpy-lab

September 19, 2025

0.1 NPS Case Study

[1]:
```python
import numpy as np
```

[3]:
```python
score = np.loadtxt('survey.txt', dtype = 'int')
```

[2]:
```python
# !gdown drive_link
```

[6]:
```python
# How do I get first 10 elements?
```

[7]:
```python
score[:10]
```

[7]:
```
array([ 7, 10,  5,  9,  9,  4,  7,  9,  9,  9])
```

[8]:
```python
# How do we filter out score wrt to detractors?
```

[9]:
```python
# Detractors are users who rated below 7
```

[12]:
```python
score < 7 # Boolean array
```

[12]:
```
array([False, False,  True, …,  True, False, False])
```

[15]:
```python
detractors = score[score < 7]
```

[16]:
```python
# NPS = % promoters - % detractors
```

[17]:
```python
# How do we find promoters
```

[20]:
```python
promoters = score[score >= 9]
```

[24]:
```python
len(detractors)
```

[24]:
```
332
```

[21]:
```python
detractors.size
```

[21]:
```
332
```

[22]:
```
promoters.size
```

[22]: 609

[23]:
```
score.size
```

[23]: 1167

[25]:
```
nps = 100 * (promoters.size - detractors.size) / score.size
```

[26]:
```
nps
```

[26]: 23.736075407026565

[27]:
```
round(nps, 2)
```

[27]: 23.74

[28]:
```
round(nps)
```

[28]: 24

[32]:
```
# detractors
```

[29]:
```
len(detractors)
```

[29]: 332

[31]:
```
sum(detractors)
```

[31]: 1080

[ ]:
```

```

[ ]:
```

```

0.2 Fitbit Case Study

[35]:
```
data = np.loadtxt('fit.txt', dtype = 'str')
```

[37]:
```
data[:10]
```

[37]:
```
array([['06-10-2017', '5464', 'Neutral', '181', '5', 'Inactive'],
       ['07-10-2017', '6041', 'Sad', '197', '8', 'Inactive'],
       ['08-10-2017', '25', 'Sad', '0', '5', 'Inactive'],
       ['09-10-2017', '5461', 'Sad', '174', '4', 'Inactive'],
       ['10-10-2017', '6915', 'Neutral', '223', '5', 'Active'],
```

```
       ['11-10-2017', '4545', 'Sad', '149', '6', 'Inactive'],
       ['12-10-2017', '4340', 'Sad', '140', '6', 'Inactive'],
       ['13-10-2017', '1230', 'Sad', '38', '7', 'Inactive'],
       ['14-10-2017', '61', 'Sad', '1', '5', 'Inactive'],
       ['15-10-2017', '1258', 'Sad', '40', '6', 'Inactive']], dtype='<U10')
```

[38]:
```
data.ndim
```

[38]: 2

[39]:
```
data.shape
```

[39]: (96, 6)

[40]:
```
# Question: What is average step count on "Active" days?
```

[42]:
```
# np.loadtxt?
```

[43]:
```
data[:10]
```

[43]:
```
array([['06-10-2017', '5464', 'Neutral', '181', '5', 'Inactive'],
       ['07-10-2017', '6041', 'Sad', '197', '8', 'Inactive'],
       ['08-10-2017', '25', 'Sad', '0', '5', 'Inactive'],
       ['09-10-2017', '5461', 'Sad', '174', '4', 'Inactive'],
       ['10-10-2017', '6915', 'Neutral', '223', '5', 'Active'],
       ['11-10-2017', '4545', 'Sad', '149', '6', 'Inactive'],
       ['12-10-2017', '4340', 'Sad', '140', '6', 'Inactive'],
       ['13-10-2017', '1230', 'Sad', '38', '7', 'Inactive'],
       ['14-10-2017', '61', 'Sad', '1', '5', 'Inactive'],
       ['15-10-2017', '1258', 'Sad', '40', '6', 'Inactive']], dtype='<U10')
```

[45]:
```
# date, step_count, mood, calories_burned, sleep_hours, status
```

[60]:
```
# We can extract data like following and by using Transpose as well:

# date = data[:, 0]
# step_count = data[:, 1]
# mood = data[:, 2]
```

[56]:
```
date, step_count, mood, calories_burned, sleep_hours, status = data.T
```

[70]:
```
step_count = step_count.astype('int')
sleep_hours = sleep_hours.astype('int')
calories_burned = calories_burned.astype('int')
```

[71]:
```
step_count
```

[71]:
```
array([5464, 6041,   25, 5461, 6915, 4545, 4340, 1230,   61, 1258, 3148,
       4687, 4732, 3519, 1580, 2822,  181, 3158, 4383, 3881, 4037,  202,
        292,  330, 2209, 4550, 4435, 4779, 1831, 2255,  539, 5464, 6041,
       4068, 4683, 4033, 6314,  614, 3149, 4005, 4880, 4136,  705,  570,
        269, 4275, 5999, 4421, 6930, 5195,  546,  493,  995, 1163, 6676,
       3608,  774, 1421, 4064, 2725, 5934, 1867, 3721, 2374, 2909, 1648,
        799, 7102, 3941, 7422,  437, 1231, 1696, 4921,  221, 6500, 3575,
       4061,  651,  753,  518, 5537, 4108, 5376, 3066,  177,   36,  299,
       1447, 2599,  702,  133,  153,  500, 2127, 2203])
```

[72]:
```
status[status == 'Active']
```

[72]:
```
array(['Active', 'Active', 'Active', 'Active', 'Active', 'Active',
       'Active', 'Active', 'Active', 'Active', 'Active', 'Active',
       'Active', 'Active', 'Active', 'Active', 'Active', 'Active',
       'Active', 'Active', 'Active', 'Active', 'Active', 'Active',
       'Active', 'Active', 'Active', 'Active', 'Active', 'Active',
       'Active', 'Active', 'Active', 'Active', 'Active', 'Active',
       'Active', 'Active', 'Active', 'Active', 'Active', 'Active'],
      dtype='<U10')
```

[73]:
```
len(status)
```

[73]:
```
96
```

[74]:
```
status == 'Active'
```

[74]:
```
array([False, False, False, False,  True, False, False, False, False,
       False, False, False,  True, False, False, False, False, False,
       False, False, False, False, False, False, False,  True, False,
       False, False, False,  True, False, False, False, False, False,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
       False, False, False, False, False, False,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
       False,  True,  True, False,  True,  True,  True,  True,  True,
       False,  True,  True,  True,  True, False, False, False, False,
        True,  True,  True,  True, False, False, False, False, False,
       False, False, False,  True, False,  True])
```

[75]:
```
step_count[status == 'Active']
```

[75]:
```
array([6915, 4732, 4550,  539, 6314,  614, 3149, 4005, 4880, 4136,  705,
        570,  269,  493,  995, 1163, 6676, 3608,  774, 1421, 4064, 2725,
       5934, 1867, 3721, 2909, 1648, 7102, 3941, 7422,  437, 1231, 4921,
        221, 6500, 3575, 5537, 4108, 5376, 3066,  500, 2203])
```

[76]:
```
np.mean(step_count[status == 'Active'])
```

[76]: 3226.5714285714284

[78]:
```python
round(np.mean(step_count[status == 'Active']), 2)
```

[78]: 3226.57

[77]:
```python
# Question: Find avg calories burned when status is inactive?
```

[79]:
```python
# Question: How many days had more than 5000 steps and burned more than 150␣
↪calories
```

[80]:
```python
step_count > 5000
```

[80]:
```
array([ True,  True, False,  True,  True, False, False, False, False,
       False, False, False, False, False, False, False, False, False,
       False, False, False, False, False, False, False, False, False,
       False, False, False, False,  True,  True, False, False, False,
        True, False, False, False, False, False, False, False, False,
       False,  True, False,  True,  True, False, False, False, False,
        True, False, False, False, False, False,  True, False, False,
       False, False, False, False,  True, False,  True, False, False,
       False, False, False,  True, False, False, False, False, False,
        True, False,  True, False, False, False, False, False, False,
       False, False, False, False, False, False])
```

[81]:
```python
calories_burned > 150
```

[81]:
```
array([ True,  True, False,  True,  True, False, False, False, False,
       False, False,  True, False, False, False, False, False, False,
       False, False, False, False, False, False, False, False, False,
        True, False, False, False,  True,  True, False,  True, False,
        True, False, False, False,  True, False, False, False, False,
       False,  True, False,  True,  True, False, False, False, False,
        True, False, False, False, False, False,  True, False, False,
       False, False, False, False,  True, False,  True, False, False,
       False,  True, False,  True, False, False, False, False, False,
        True, False,  True, False, False, False, False, False, False,
       False, False, False, False, False, False])
```

[82]:
```python
(step_count > 5000) & (calories_burned > 150)
```

[82]:
```
array([ True,  True, False,  True,  True, False, False, False, False,
       False, False, False, False, False, False, False, False, False,
       False, False, False, False, False, False, False, False, False,
       False, False, False, False,  True,  True, False, False, False,
        True, False, False, False, False, False, False, False, False,
       False,  True, False,  True,  True, False, False, False, False,
```

```
        True, False, False, False, False, False,  True, False, False,
       False, False, False, False,  True, False,  True, False, False,
       False, False, False,  True, False, False, False, False, False,
        True, False,  True, False, False, False, False, False, False,
       False, False, False, False, False, False])
```

[85]:
```python
np.sum((step_count > 5000) & (calories_burned > 150))
```

[85]: 17

[86]:
```python
len((step_count > 5000) & (calories_burned > 150))
```

[86]: 96

[89]:
```python
# What number of days had a "Sad" mood?
# Find percentage as well?
```

[90]:
```python
# What is the maximum number of steps taken on days with less than 6 hours of␣
↪sleep?
```

[91]:
```python
# Which mood had the highest average step count?
```

[ ]:
```python

```

[92]:
```python
# Is there a correlation in number of steps and the mood a person has?
```

[94]:
```python
np.unique(mood)
```

[94]:
```
array(['Happy', 'Neutral', 'Sad'], dtype='<U10')
```

[97]:
```python
# Avg steps wrt to different moods?
```

[ ]:
```python

```

[99]:
```python
# When mood is Happy

np.mean(step_count[mood == 'Happy'])
```

[99]: 3392.725

[101]:
```python
step_count[mood == 'Happy'].mean()
```

[101]: 3392.725

[104]:
```python
# When mood is Neutral
```

[102]:
```python
np.mean(step_count[mood == 'Neutral'])
```

[102]: 3153.777777777778

[103]:
```python
# When mood is Sad

np.mean(step_count[mood == 'Sad'])
```

[103]: 2103.0689655172414

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]: